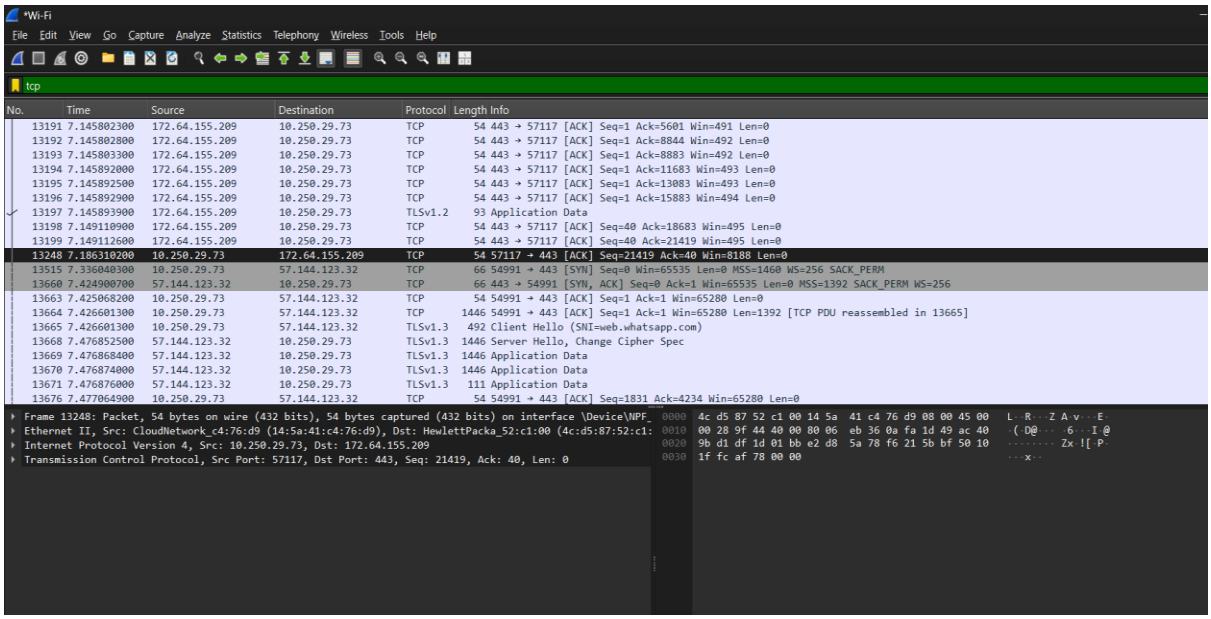
Skill -1

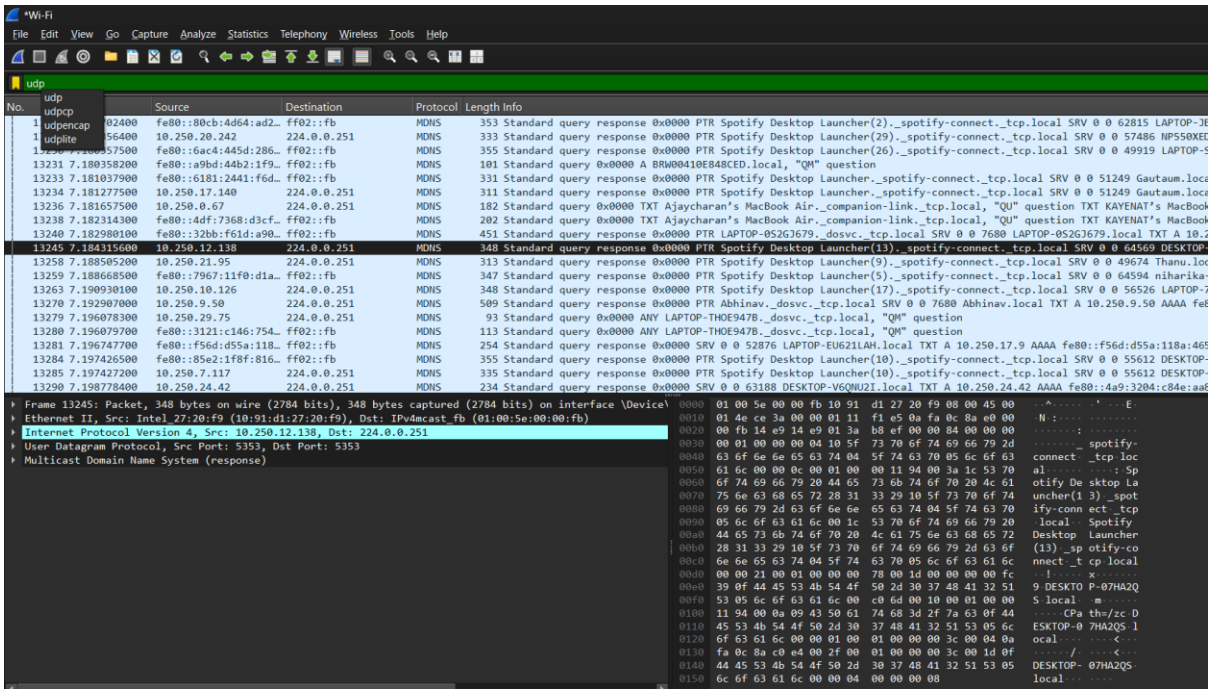
2520090235, V Rohita

Task-1

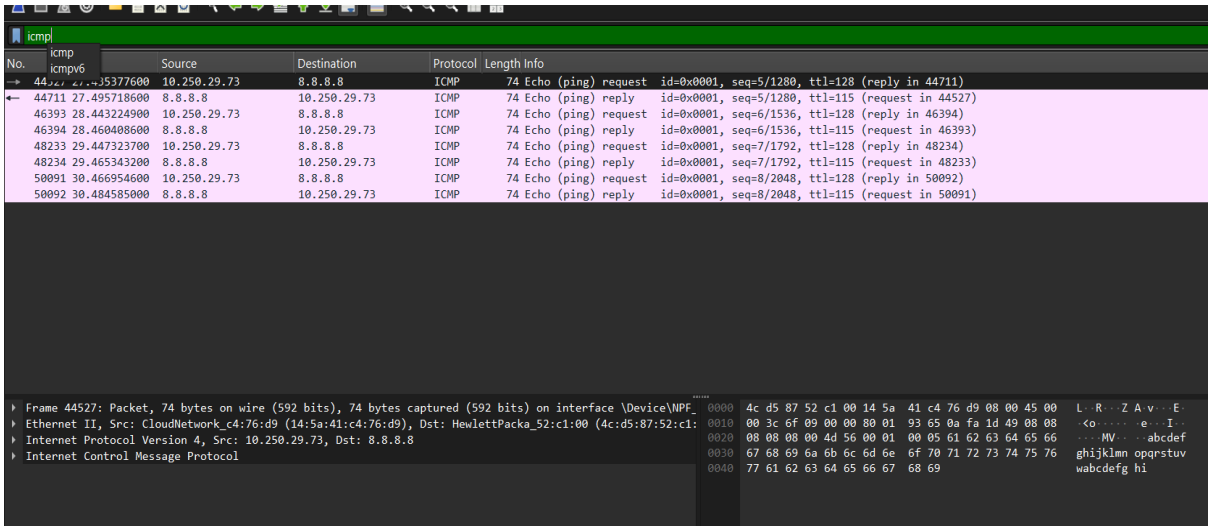
Apply the filter tcp and record the number of TCP packets.



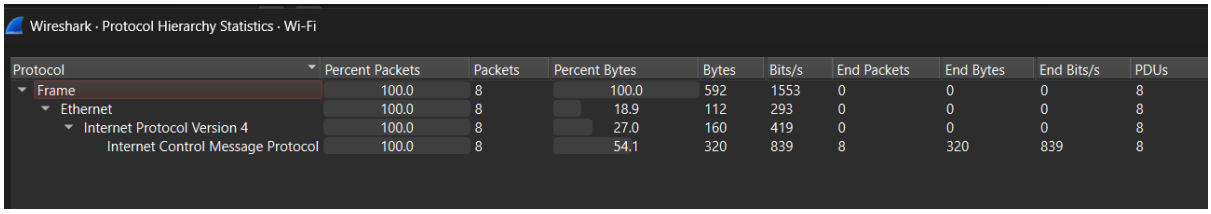
Apply the filter udp and record the number of UDP packets.



Apply the filter icmp and record the number of ICMP packets.

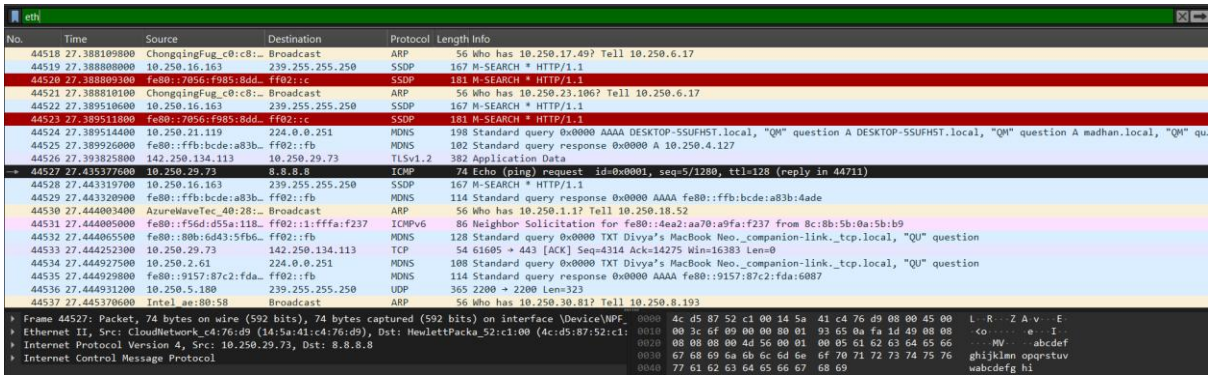


Protocol- Hierarchy



Task-2

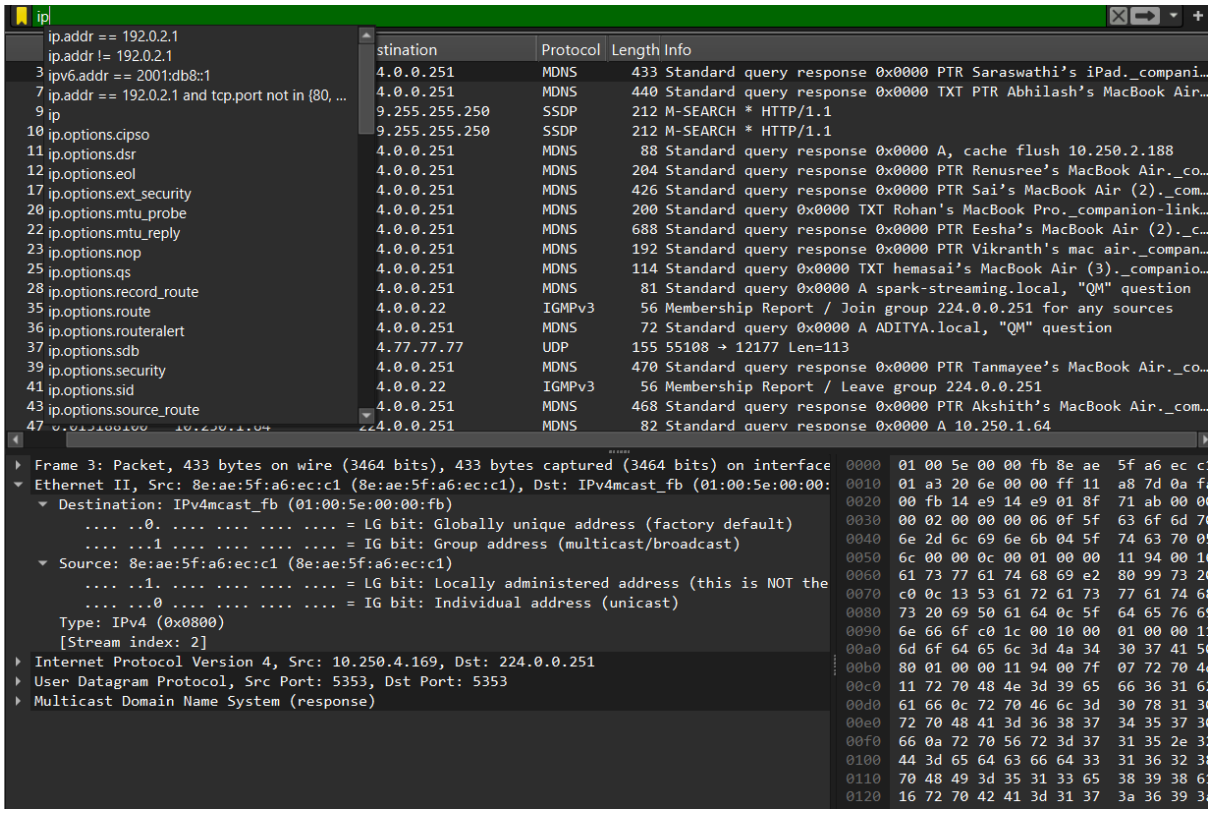
Apply the filter eth



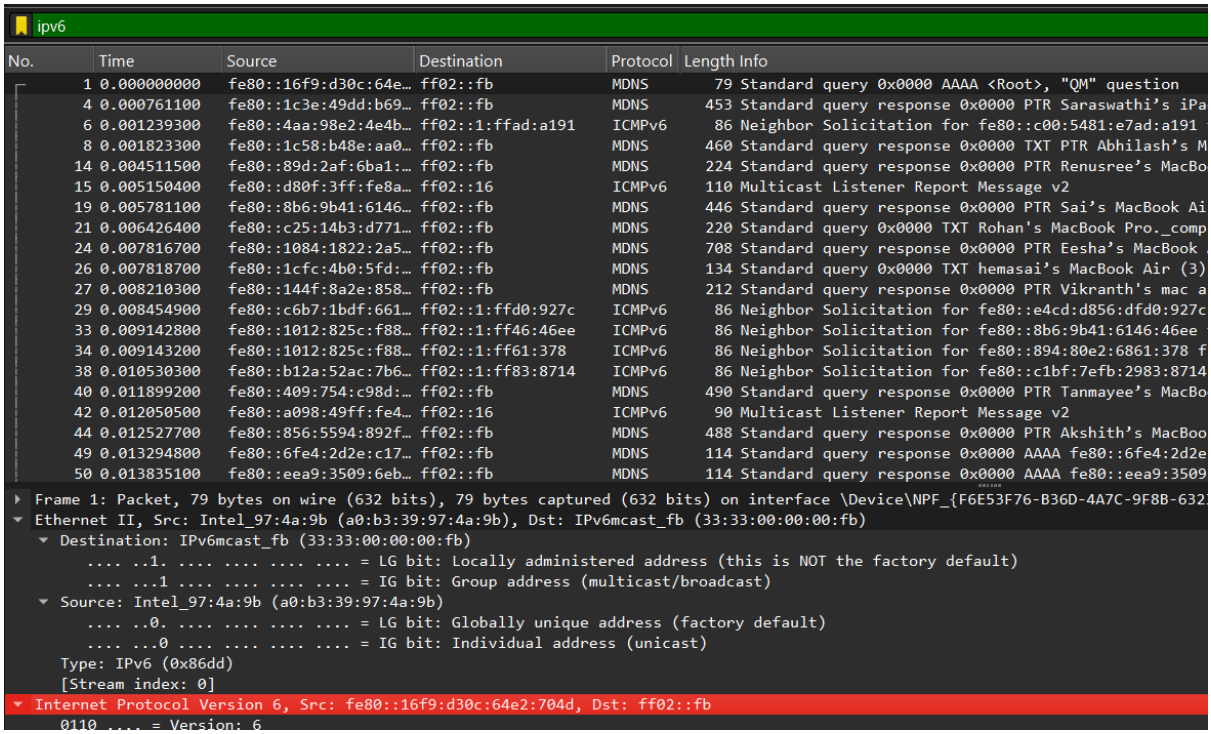
No.	Time	Source	Destination	Protocol	Length
1	0.000000000	fe80::16f9:d30c:64e…	ff02::fb	MDNS	79
2	0.000001300	4a:5a:3b:61:70:fc	Broadcast	ARP	56
3	0.000565800	10.250.4.169	224.0.0.251	MDNS	433
4	0.000761100	fe80::1c3e:49dd:b69…	ff02::fb	MDNS	453
5	0.001209600	Intel_8a:b5:60	Broadcast	ARP	56
6	0.001239300	fe80::4aa:98e2:4e4b…	ff02::1:ffad:a191	ICMPv6	86
7	0.001822500	10.250.7.217	224.0.0.251	MDNS	440
8	0.001823300	fe80::1c58:b48e:aa0…	ff02::fb	MDNS	460
9	0.002455800	10.250.11.53	239.255.255.250	SSDP	212
10	0.003903900	10.250.11.53	239.255.255.250	SSDP	212
11	0.003904600	10.250.2.188	224.0.0.251	MDNS	88
12	0.003905700	10.250.4.233	224.0.0.251	MDNS	204

Frame number, source, destination , length

Ip4

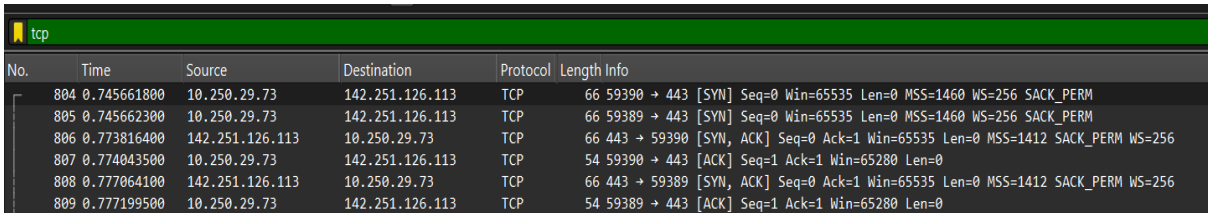


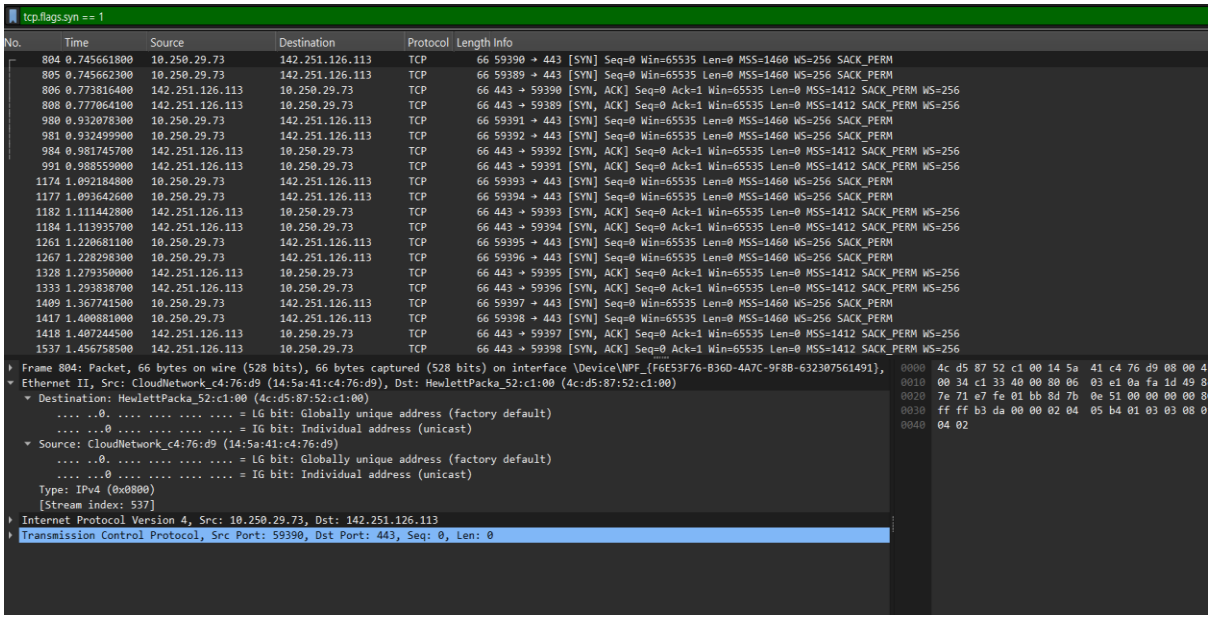
Ip version 6

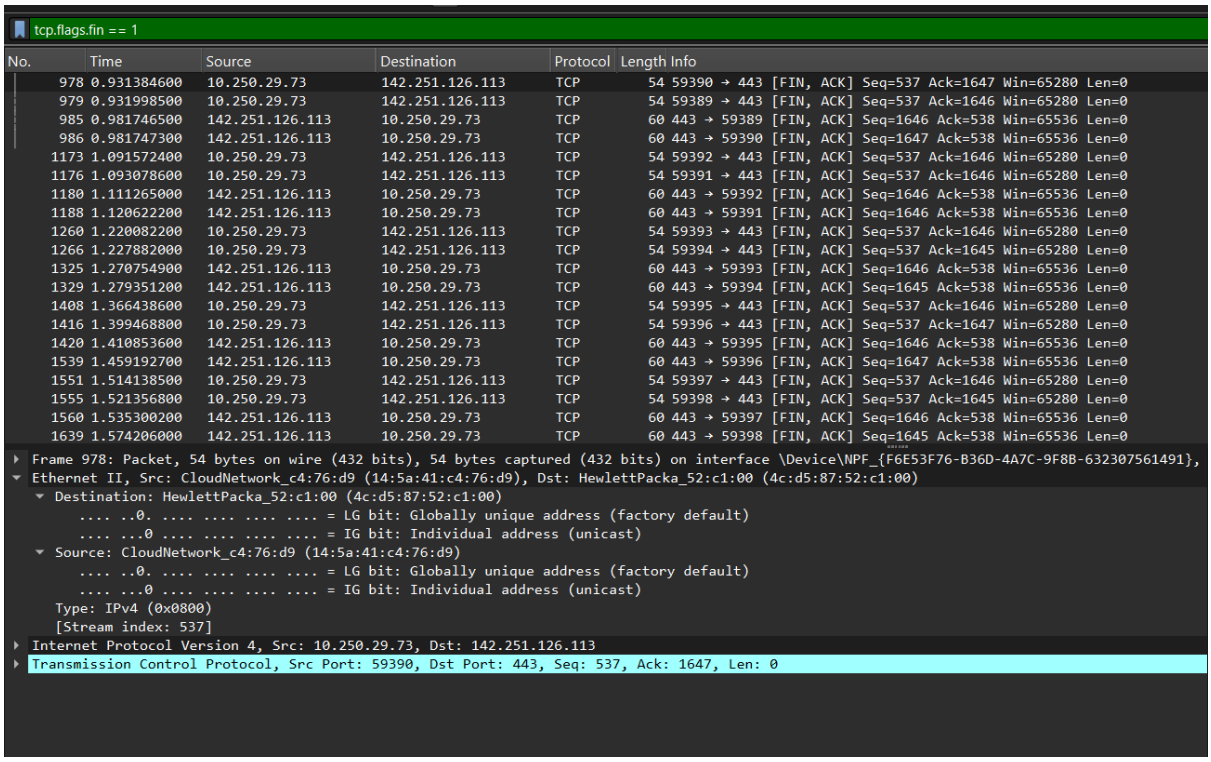


Task4- TCP Header inspection

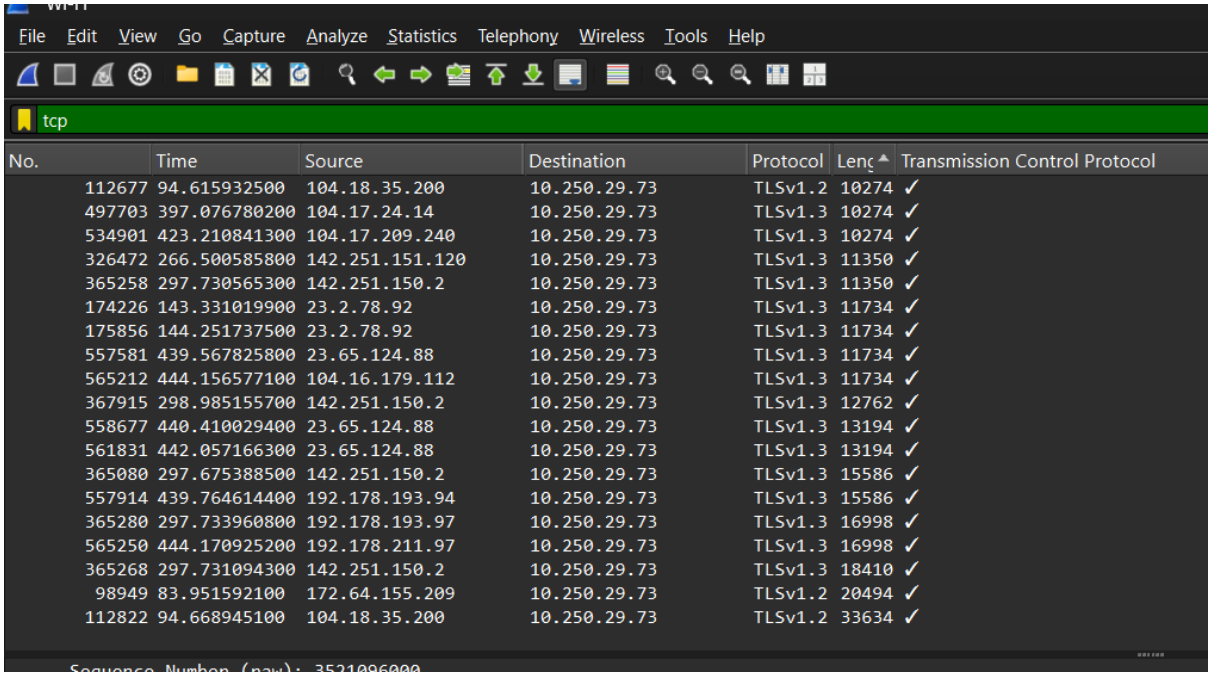
Apply TCP filter



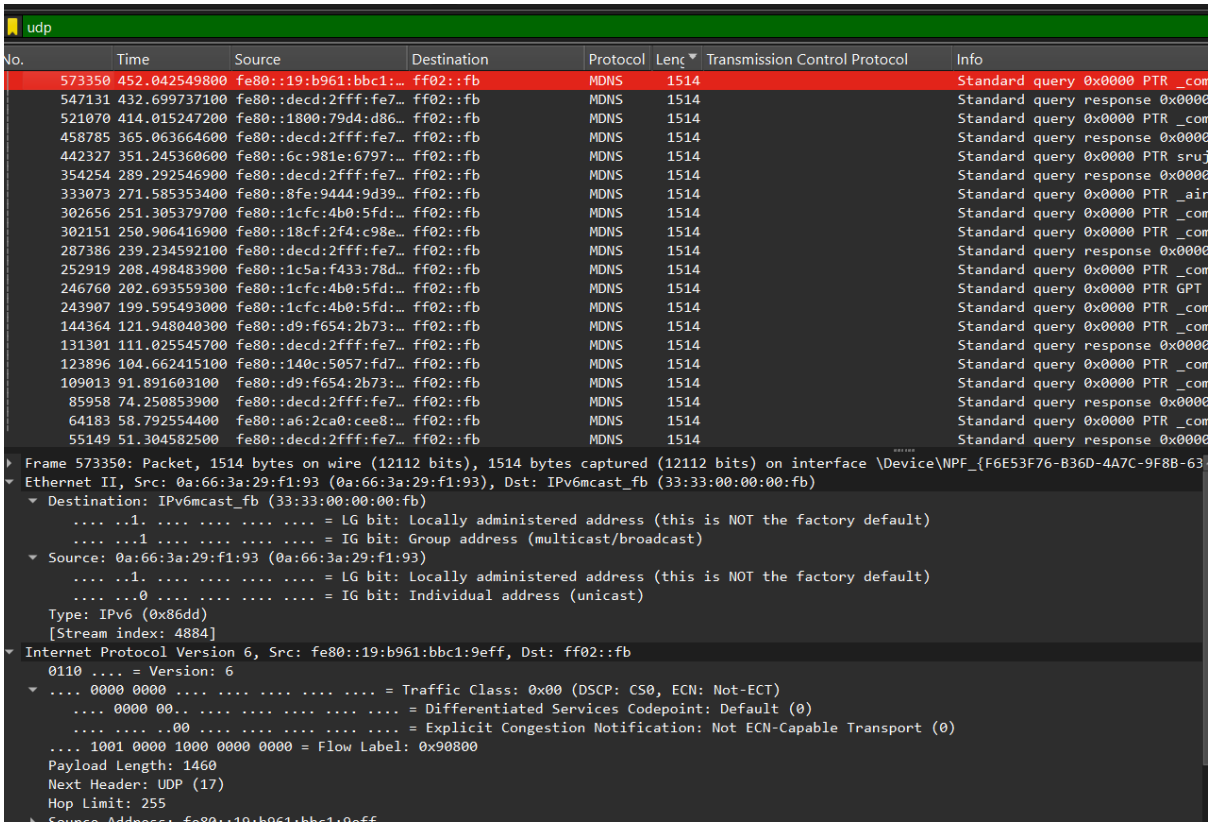




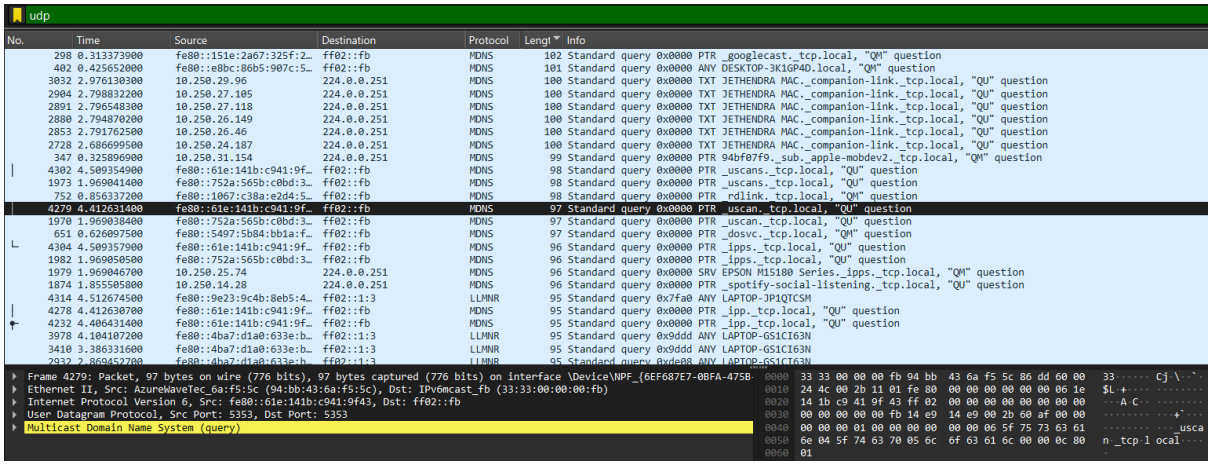
TCP packet with the maximum payload length



UDP packet with the maximum payload length.



Task 5: UDP header inspection



Select five UDP packets and record:

S.No.	Frame No.	Source Port	Destination Port	UDP Length	Payload Length
1	4279	5353	5353	43 bytes	35 bytes
2	2904	5353	5353	66 bytes	58 bytes
3	347	5353	5353	65 bytes	57 bytes
4	4278	5353	5353	41 bytes	33 bytes
5	2853	5353	5353	66 bytes	58 ytes

Identify the UDP packet with the maximum payload length.

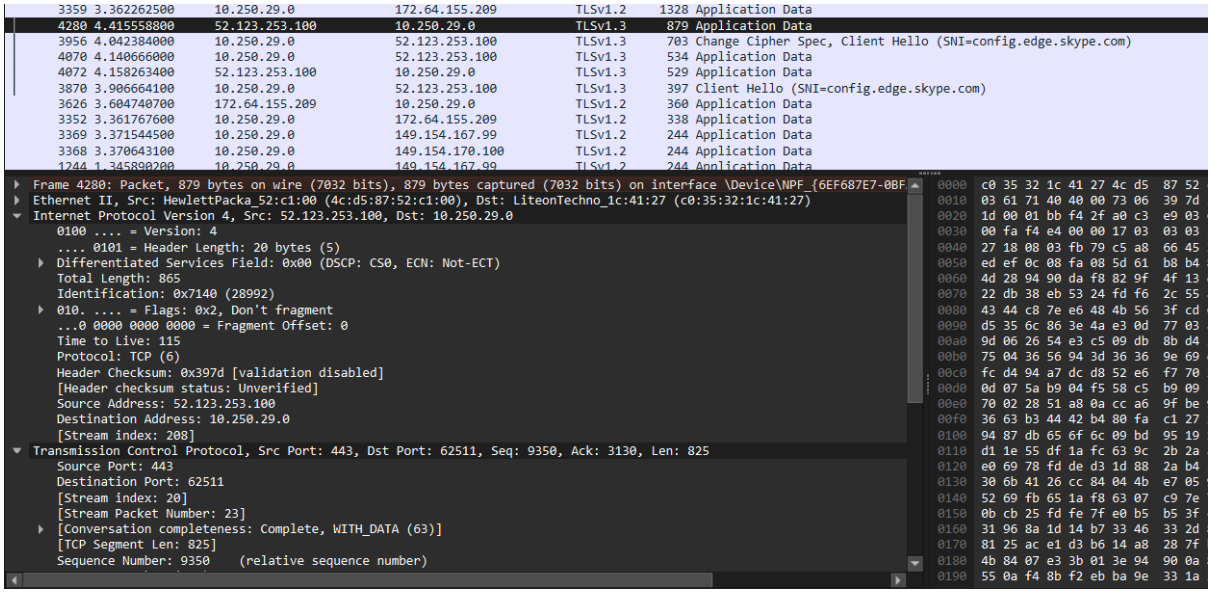
From the selected 5 Frames, Frame Number 2904 has Max Payload Length of 58 bytes.

Find the UDP packets carrying DNS traffic using the filter



Task 6: Encapsulation and overhead calculation

Select one TCP packet containing application data. Record the Ethernet, IPv4 and TCP header lengths.



Protocol	Header Length
Ethernet II	14 bytes
IPv4	20 bytes
TCP	20 bytes

Calculate the total protocol header overhead for the selected packet.

Formula:

Total Overhead = Ethernet + IPv4 + TCP

= 14 + 20 + 20 = 54 bytes

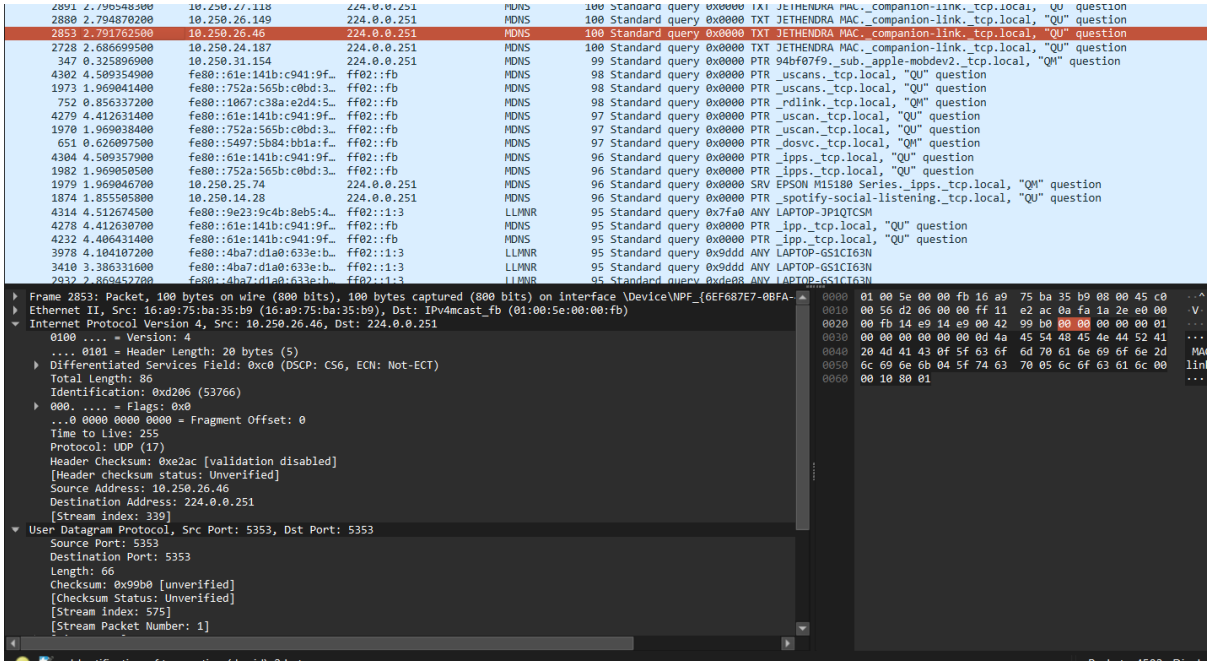
Calculate the payload efficiency percentage.

Formula:

*Efficiency = (Payload/Total Frame Length) * 100*

= (825/879) * 100 = 93.86%

Repeat the same calculations for one UDP packet.



Total Overhead = 14+20+8 = 42 bytes

Payload = 66-8 = 58

Efficiency = (58/100) * 100 = 58%